

Q4(a)

$$E = V + Ir$$

$$E = 5.66 + 0.724r \dots\dots(1)$$

$$E = 5.96 + 0.405r \dots\dots(2)$$

$$\text{Hence } 5.66 + 0.724r = 5.96 + 0.405r$$

$$\mathbf{r \approx 0.94 \, \Omega}$$

$$\text{In (1), } E = 5.66 + 0.724(0.94)$$

$$\mathbf{E \approx 6.34 \, V}$$

Q4(b)

Use the terminal p.d. and the circuit current:

$$R_{eff} = \frac{V}{I} = \frac{5.96}{0.405}$$

$$\therefore R_{eff} = \mathbf{14.7 \, \Omega}.$$

Q4(c)

Use the formula for resistors connected in parallel.

$$\frac{1}{R_{eff}} = \frac{1}{R} + \frac{1}{R_v}$$

When $x = 100.0$ cm,

$$\frac{1}{14.7} = \frac{1}{R} + \frac{1}{R_v} \quad - (1)$$

When $x = 50.0$ cm,

$$R_{eff} = \frac{V}{I} = \frac{5.66}{0.724} = 7.82 \, \Omega$$

$$\therefore \frac{1}{7.82} = \frac{1}{0.50R} + \frac{1}{R_v} \quad - (2)$$

Substitute $1/R_v$ from (1) into (2):

$$\frac{1}{7.82} = \frac{1}{0.50R} + \frac{1}{14.7} - \frac{1}{R}$$

$$\therefore \frac{1}{R} = \frac{1}{7.82} - \frac{1}{14.7}$$

$$\therefore R = \mathbf{16.7 \, \Omega}$$

Substitute into (1), we get:

$$R_v = \mathbf{123 \, \Omega}$$

(d) At the balance point,

e.m.f. of the 1st cell = voltage across XJ

$$E = V_{XJ}$$

$$E = \frac{R_{XJ}}{R} \times E_c$$

$$\therefore 6.34 = 0.365 \times E_c$$

$$\therefore E_c = \mathbf{17.4 \, V}$$